



Infection Prevention and Management Manual

Hand Hygiene

Scope

Site	Service/Department/Unit	Disciplines
SCGHOPHCG	Organisation Wide	All Staff

Table of Contents

Introduction	2
Risk Statement	2
Definitions	2
Policy Principles.....	2
1.0 Responsibilities.....	3
2.0 Indications for Hand Hygiene.....	3
3.0 Procedure	3
3.1 Routine handwashing.....	3
3.2 Alcohol-based hand rub (ABHR)	4
3.3 Antiseptic handwashing.....	4
3.4 Surgical handwashing	5
3.5 Alcohol-based Surgical hand rub	5
4.0 Hand Hygiene for Patients and Visitors	5
5.0 Hand care	5
6.0 Healthcare workers with exfoliative skin conditions	6
7.0 Healthcare workers wearing hand or wrist splints.....	6
8.0 Bare below elbows.....	6
9.0 Glove use	6
10.0 Hand hygiene stations and dispensers	7
Compliance, monitoring and evaluation.....	7
Related Documents	8
References	8
Authorisation & Version Control.....	8
Appendix 1: How to Handwash?.....	9
Appendix 2: How to Handrub?	10
Appendix 3: Surgical Handrubbing Technique.....	11

Introduction

Healthcare worker hands are a major source of transmission of micro-organisms. Hand hygiene results in the reduction of carriage of potential pathogens on hands. Applying the principals for hand hygiene has been demonstrated as being the most effective and efficient means to reduce the risk, to both patients and staff, of healthcare-associated infection.

Risk Statement

Non-compliance with this policy will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☒
Other:	☐		

Definitions

Hand Hygiene	Term that applies to the processes of either handwashing or hand decontamination, to reduce the number of microorganisms on hands.
Handwashing	Process that involves the mechanical removal of bacteria from hands with plain non-antimicrobial soap and water.
Hand decontamination	Process of removing or reducing the number of viable bacteria from hands with antimicrobial soap and water or using alcohol-based hand rub (ABHR).
Healthcare Worker	For the purposes of this policy, a healthcare worker (HCW) is deemed to be any staff member listed below who has direct contact with patients, their belongings or the patient zone: • Clinical staff e.g., Doctors, Nurses, Assistants in Nursing, Allied Health, Medical Imaging, and PathWest staff) • Patient Support Services • Clerical/administration staff • Maintenance, contractors and security staff • Volunteers
Healthcare zone	The healthcare zone includes all regions outside of the patient zone. This includes curtains, partitions and doors between separate patient areas.
Patient equipment	For the purposes of this policy, patient equipment refers to all clinical equipment, patient furniture and the patient medical record.
Patient Zone	The space dedicated to an individual patient for the patients stay. The patient zone includes the patient's immediate surroundings, but does not include curtains/privacy screens, doors and partitions that separate patient areas. This area is cleaned after one patient leaves and before the next patient arrives.

Policy Principles

The purpose of this policy is to provide direction to all staff on the expected standard for hand hygiene across the Sir Charles Gairdner Osborne Park Health Care Group (SCGOPHCG).

1.0 Responsibilities

All staff are responsible for ensuring they perform hand hygiene correctly and encourage others delivering care, visitors, and patients to perform hand hygiene.

The SCGOPHCG Infection Prevention and Management Committee is responsible for implementing a Hand Hygiene Program that is consistent with the Australian Safety and Quality Health Service Standards for Hand Hygiene, ensuring:

- Hand hygiene compliance audits are submitted to the National Hand Hygiene Initiative (NHII) in accordance with NMHS and national data submission requirements
- Results of audits are provided to staff/departments and executive to encourage improvement and sustain high levels of compliance.

2.0 Indications for Hand Hygiene

A hand hygiene moment is where there is a perceived or actual risk of organism transmission from one surface to another via hands when healthcare workers are in the patient or healthcare zones. There are **Five Moments for hand hygiene** that have been identified as being critical for hand hygiene performance.

These 5 moments are:

1. Before touching a patient
2. Before a procedure
3. After a procedure or body fluid exposure risk
4. After touching a patient
5. After touching a patient's surroundings

In addition, hand hygiene should be performed:

- Between different care activities for the same patient
- Before and after removing personal protective equipment, e.g., gloves, aprons, masks
- Before eating, handling food or dispensing medication
- To maintain personal hygiene
- After blowing your nose or after coughing/sneezing into a tissue
- When performing maintenance duties in the facility including environmental hygiene and waste management
- After using a computer keyboard or mobile device in clinical/patient areas
- Before handling linen

3.0 Procedure

3.1 Routine handwashing

Aim	To remove dirt, organic material, and transient microorganisms.
Indications	When hands are visibly soiled or potentially contaminated with dirt or organic material. Hand hygiene must be performed using soap and water when <i>Clostridioides difficile</i> (<i>Clostridium difficile</i>) or gastroenteritis viruses such as norovirus, are known or suspected and gloves have not been worn.
Product	Liquid soap/foam hand wash solution, non-medicated, neutral pH.

Procedure as indicated in Appendix 1: How to Handwash?	• Ensure bare below the elbows • Apply liquid soap/foam handwash solution to wet hands • Rub hands and wrist, and lather vigorously for at least 20 seconds paying particular attention to the tops of the fingers, thumbs, and the areas in between the fingers • Avoid contact with taps. If elbow or foot controls are not available, use paper towel to turn off taps • Pat hands dry with disposable paper towels
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3.2 Alcohol-based hand rub (ABHR)

Aim:	To reduce and inhibit the number of viable micro-organisms on hands. When hands are visibly clean, ABHR is more effective against most bacteria and many viruses than washing with plain soap and water.
Indications	• Use on visibly clean and dry hands only: - ABHR will not remove physical dirt or soiling - ABHR applied to wet hands is vulnerable to dilution/inactivation • Preferred hand hygiene product for all standard aseptic technique procedures. • Not suitable for use when hands are visibly soiled or where gloves have not been worn and the patient has known or suspected <i>Clostridioides difficile</i>.
Product	Alcohol-based hand rub.
Procedure as indicated in Appendix 2: How to Handrub?	• Ensure bare below the elbows • Hands are to be visibly clean and dry. • Apply 1-2 pumps of hand rub to hands • Rub hands and wrists together for at least 20 seconds until dry.

NOTE: There is no maximum number of times that ABHR can be used before hands need to be washed with soap and water.

3.3 Antiseptic handwashing

Aim	To reduce and inhibit the number of viable organisms on hands.
Indications	An alternative to ABHR prior to invasive or diagnostic procedures e.g., insertion of central venous catheters, sterile procedures such as lumbar puncture.
Product	Chlorhexidine gluconate 4% or 2% chlorhexidine antimicrobial foaming hand wash.
Procedure as indicated in Appendix 1: How to Handwash?	Wash hands thoroughly using an antimicrobial soap for two minutes using the technique outlined in Routine Handwashing.

3.4 Surgical handwashing

Aim	To achieve antisepsis by removing transient microorganisms and reducing the resident flora.
Indications	Prior to all surgical procedures or procedures requiring critical asepsis.
Product	Antimicrobial soap or an alcohol/chlorhexidine-based hand rub with persistent antimicrobial activity. Staff who are required to perform a surgical hand wash and are unable to use chlorhexidine or iodine must seek advice from the Department of Health, Safety and Wellness.
Procedure	Refer to area-specific policies and guidelines.

3.5 Alcohol-based Surgical hand rub

Aim	To achieve antisepsis by reducing and inhibiting the number of viable organisms on the hands and forearms for the duration of a surgical procedure
Indications	Prior to all surgical procedures or procedures requiring critical asepsis. If hands are visibly soiled or dirty, wash hands with non-antimicrobial soap and water, and ensure hands are dry before applying surgical hand rub
Product:	Alcohol-based surgical hand rub.
Procedure as indicated in Appendix 3: Surgical Handrubbing Technique .	Follow manufacturers' instructions regarding recommended application times and volume of product to be used. Do not combine surgical hand scrub and alcohol-based surgical hand rub sequentially as it may result in moist hands and dilution of the antimicrobial properties of the surgical hand rub as well as enhance skin irritation and dryness.

4.0 Hand Hygiene for Patients and Visitors

It is important for both patients and visitors to perform hand hygiene.

Hand hygiene facilities should be readily available for each patient, and education and assistance should be provided where necessary. Patients should be encouraged to perform hand hygiene prior to eating, after using the toilet/bedpan and on entering and leaving their room.

Visitors should perform hand hygiene on entry and upon leaving the patient's room.

5.0 Hand care

Intact skin is a natural defence against infection. Healthcare workers with cuts or abrasions should cover them with a waterproof dressing, e.g., transparent occlusive dressing.

Moisturising hand lotion, compatible with the hand hygiene/antiseptic products is supplied by the hospital and should be applied regularly, for example at meal breaks and on completion of the shift to prevent dryness.

6.0 Healthcare workers with exfoliative skin conditions

Healthcare workers (HCWs) who have an exfoliative skin condition or hand dermatitis must report to the Health, Safety and Wellbeing Department for advice.

7.0 Healthcare workers wearing hand or wrist splints

HCWs who work in a clinical area and wear a hand or wrist splint must report to the Health, Safety and Wellbeing Department to determine their ability to perform effective hand hygiene when providing clinical care.

8.0 Bare below elbows

Forearms must be kept bare below the elbows to ensure that handwashing is effective. The effectiveness of hand hygiene is improved when skin is intact; nails are natural, short and unvarnished; hands and forearms are free of jewellery; and sleeves are above the elbow.

All staff who may have contact with patient belongings, patient equipment, the patient zone, as well as other frequently used communal items in the clinical areas e.g. medical records, keyboards, phones and touch screen devices, shall comply with the following;

- Bracelets and wrist watches (including Fitbits and smart watches) are not to be worn
- Rings with stones or ridges are not to be worn. A single flat ring/band is permitted unless working in a high-risk area i.e., theatre, where no rings are acceptable
- Medical alert necklace is preferred to a medical alert bracelet
- Long sleeved clothing should be rolled up above the elbow (does not apply to the use of long-sleeved gown when indicated for Standard and Transmission-based precautions)
- Attire that is not laundered daily (e.g., cardigans/jumpers) must not be worn whilst providing direct patient care/services
- Fingernails shall be kept short ($\leq 0.5\text{cm}$ long), clean and free of nail polish
- Nail polish and artificial nails or extenders must not be worn in the clinical setting.

NOTE: Artificial/nail polish refers to all types of nail coverings, including, but not limited to, polish, shellac, Signature Nail System (SNS) and acrylic nails.

9.0 Glove use

Inappropriate glove use undermines the importance of hand hygiene, increases the risk of organism transmission, generates unnecessary waste and is a wasted resource.

Wearing gloves does not replace the need for hand hygiene. Gloves do not provide complete protection against hand contamination. Pathogens can contaminate healthcare workers' hands via small defects in gloves, or by contamination of the hands during glove removal. Additionally, prolonged, and indiscriminate use of gloves will increase the risk of adverse reactions and skin sensitivity.

In principle, gloves are only required:

- For contact with blood or other potentially infectious materials, mucous membranes

- or non-intact skin
- As indicated for transmission-based precautions
- For surgical aseptic technique procedures and contact with sterile sites.

Hand hygiene must be performed:

- Prior to donning gloves
- Immediately following removal. Gloves do not always provide an impermeable barrier and there is a risk of hand contamination during doffing procedures.

Gloves should be removed and hand hygiene performed;

- Between episodes of care for different patients
- According the 5 moments for HH moments during the care of a single patient to prevent cross-contamination between body sites
- If the patient interaction involves touching mobile equipment that is being used between patients.

Gloves are always single use. Hand hygiene, including using ABHR must not be undertaken while wearing gloves. Gloves must be discarded immediately following removal.

10.0 Hand hygiene stations and dispensers

ABHR is to be readily available in patient-care areas, preparation, and treatment rooms, and at the point of care. Hand hygiene pump bottles must not be refilled and should be discarded when empty.

The use of nailbrushes is not recommended unless performing surgical handwashing. If a nailbrush is required to remove heavy soiling it must be sterile and discarded after use.

All clinical and procedural handwash basins must be kept clean and free from items and equipment such as flowers and toiletries.

All clinical and procedural handwash basins must be used for handwashing only.

- Do not discard drinks/fluids for consumption, antibiotics, body fluids or water from patient wash bowls into hand wash basins
- Do not wash dishes or equipment in the clinical / procedural hand wash basin
- Do not store equipment or patients' personal items on or within one metre of the handwash basin.

Brackets and dispensers holding hand hygiene products must be kept clean and in good working condition.

Broken / damaged brackets and dispensers are to be notified for repair or replacement i.e., submit an EMPAC.

Compliance, monitoring and evaluation

Compliance against this policy will be coordinated and evaluated by the Infection Prevention and Management Unit and SCGOPHCG Infection Prevention and Management Committee via:

- Hand hygiene auditing in accordance with requirements of the National Hand Hygiene Initiative (NHHI)
- 'Bare below the elbows' auditing
- Environmental auditing, e.g., hand hygiene station maintenance and management.

Related Documents

- NSQHS Standard 3 - Preventing and Controlling Healthcare-Associated Infection
- NMHS Policy 158. Dress Code for Staff.

References

1. Australian College of Operating Room Nurses. 2023 Professional Practice Standards for Perioperative Nurses (PPSPNs) for Organisations. 2023 [cited 2023 Sep 8]. Available from: <https://www-webdoxx.com.qelibresources.health.wa.gov.au/acorn/acorn15/index.php?page=1&loggedout=1>
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3. Australian Government National Health and Medical Research Council. Australian Guidelines for the Prevention and Control of Infection in Healthcare [Internet]. 2019 [cited 2020 Oct 27]. Available from: <https://www.nhmrc.gov.au/about-us/publications/australian-guidelines-prevention-and-control-infection-healthcare-2019>
4. World Health Organisation. WHO guidelines on hand hygiene in healthcare: First global patient safety challenge clean care is safer care [Internet]. 2009 [cited 2020 Oct 14]. Available from: <https://www.who.int/gpsc/5may/tools/9789241597906/en/>

Authorisation & Version Control

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Policy Contact	Coordinator of Nursing, Infection Prevention and Management				
Date First Issued:	01/01/1986	Last Reviewed:	October 2023	Review Date:	October 2026
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Standards Applicable:	Insert relevant NSQHS Standards (Version 2): 				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1: How to Handwash?

Diagram from: https://www.who.int/gpsc/5may/How_To_HandWash_Poster.pdf?ua=1

How to Handwash?

WASH HANDS WHEN VISIBLY SOILED! OTHERWISE, USE HANDRUB



Duration of the handwash (steps 2-7): 15-20 seconds



Duration of the entire procedure: 40-60 seconds

0



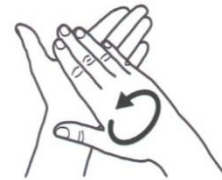
Wet hands with water;

1



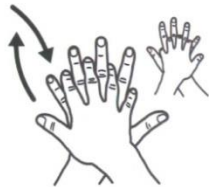
Apply enough soap to cover all hand surfaces;

2



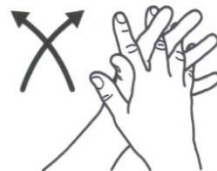
Rub hands palm to palm;

3



Right palm over left dorsum with interlaced fingers and vice versa;

4



Palm to palm with fingers interlaced;

5



Backs of fingers to opposing palms with fingers interlocked;

6



Rotational rubbing of left thumb clasped in right palm and vice versa;

7



Rotational rubbing, backwards and forwards with clasped fingers of right hand in left palm and vice versa;

8



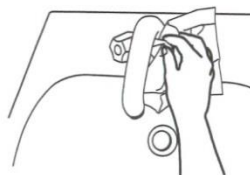
Rinse hands with water;

9



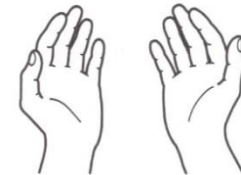
Dry hands thoroughly with a single use towel;

10



Use towel to turn off faucet;

11



Your hands are now safe.



World Health Organization

Patient Safety

A World Alliance for Safer Health Care

SAVE LIVES

Clean Your Hands

Based on the 'How to Handwash', URL: http://www.who.int/gpsc/5may/How_To_HandWash_Poster.pdf © World Health Organization 2009. All rights reserved

Appendix 2: How to Handrub?

Diagram from: https://www.who.int/gpsc/5may/How_To_HandRub_Poster.pdf?ua=1

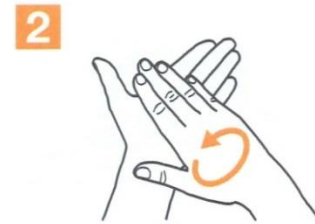
How to Handrub?

RUB HANDS FOR HAND HYGIENE! WASH HANDS WHEN VISIBLY SOILED

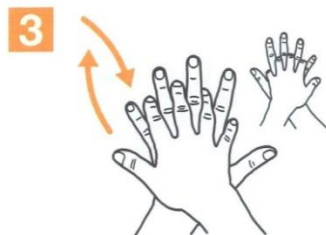
Duration of the entire procedure: 20-30 seconds



Apply a palmful of the product in a cupped hand, covering all surfaces;



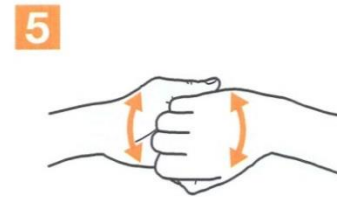
Rub hands palm to palm;



Right palm over left dorsum with interlaced fingers and vice versa;



Palm to palm with fingers interlaced;



Backs of fingers to opposing palms with fingers interlocked;



Rotational rubbing of left thumb clasped in right palm and vice versa;



Rotational rubbing, backwards and forwards with clasped fingers of right hand in left palm and vice versa;



Once dry, your hands are safe.



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SAVE LIVES
Clean Your Hands

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Appendix 3: Surgical Handrubbing Technique

Diagram from: [https://cdn.who.int/media/docs/default-source/integrated-health-services-\(ihs\)/ssi/poster/hh-surgicala3.pdf?sfvrsn=508b7e60_2](https://cdn.who.int/media/docs/default-source/integrated-health-services-(ihs)/ssi/poster/hh-surgicala3.pdf?sfvrsn=508b7e60_2)

Surgical Handrubbing Technique

- Handwash with soap and water on arrival to OR, after having donned theatre clothing (cap/hat/bonnet and mask).
- Use an alcohol-based handrub (ABHR) product for surgical hand preparation, by carefully following the technique illustrated in Images 1 to 17, before every surgical procedure.
- If any residual talc or biological fluids are present when gloves are removed following the operation, handwash with soap and water.



1 Put approximately 5ml (3 doses) of ABHR in the palm of your left hand, using the elbow of your other arm to operate the dispenser.



2 Dip the fingertip of your right hand in the handrub to decontaminate under the nails (5 seconds).



Images 3-7: Smear the handrub on the right forearm up to the elbow. Ensure that the whole skin area is covered by using circular movements around the forearm until the handrub has fully evaporated (10-15 seconds).



Images 8-10: Now repeat steps 1-7 for the left hand and forearm.

Put approximately 5ml (3 doses) of ABHR in the palm of your left hand as illustrated, to rub both hands at the same time up to the wrists, following all steps in images 12-17 (20-30 seconds).

Cover the whole surface of the hands up to the wrist with ABHR, rubbing palm against palm with a rotating movement.



13 Rub the back of the left hand, including the wrist, moving the right palm back and forth, and vice-versa.



14 Rub palm against palm back and forth with fingers interlinked.



15 Rub the back of the fingers by holding them in the palm of the other hand with a sideways back and forth movement.



16 Rub the thumb of the left hand by rotating it in the clasped palm of the right hand and vice versa.



17 When the hands are dry, sterile surgical clothing and gloves can be donned.

Repeat this sequence (average 60 sec) the number of times that adds up to the total duration recommended by the ABHR manufacturer's instructions. This could be two or even three times.



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